Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/30/2021
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Does Intermittent or Continuous Artificial Gravity Counteract Long Duration Bed Rest Induced Neurocognitive Declines?		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rachaelseidler@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-294-1722
Organization Name:	University of Florida		
PI Address 1:	Applied Physiology & Kinesiology		
PI Address 2:	FLG 142, P.O. Box 118205		
PI Web Page:			
City:	Gainesville	**State:**	FL
Zip Code:	32611-8205	**Congressional District:**	3
Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-AGBR. Appendix G: Physiological & Behavioral Responses in Humans to Intermittent Artificial Gravity during Bed Rest
Start Date:	07/01/2018	**End Date:**	06/30/2022
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2022 per NSSC information (Ed., 6/18/21) NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 8/28/20)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Mulavara, Ajitkumar Ph.D. (KBR/NASA Johnson Space Center) Wood, Scott Ph.D. (NASA Johnson Space Center) Bloomberg, Jacob Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC18K0783		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Research Announcement NNJ15ZSA001N-AGBR: Appendix G: Physiological and Behavioral Responses in Humans to Intermittent Artificial Gravity during Bed Rest invites proposals to investigate whether 30 minutes of artificial gravity (AG) per day, delivered either continuously or in six, five minute intermittent blocks, mitigates changes that occur in a spaceflight analog environment. We have recently completed a 70 day 6° head down tilt bed rest study in which we identified specific bed rest induced changes in brain structure, brain function, sensorimotor, and cognitive performance. Here, we select those that are sensitive to this spaceflight analog environment as outcome metrics for the following aims: Aim 1) Determine whether 30 minutes per day of supine centrifugation +1Gz at the heart is effective at mitigating neurocognitive declines induced by 60 days of 6° head down tilt bed rest in comparison to a control group that is not exposed to AG. Aim 2) Determine whether centrifugation is more effective and better tolerated as a countermeasure when it is delivered continuously for 30 minutes or in six, five minute sessions separated by breaks of five minutes each. Our approach utilizes cutting edge neuroimaging techniques and a broad ranging battery of sensory, motor, and cognitive assessments that we have found to be sensitive indices of bed rest—and in several cases—spaceflight induced performance decrements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research on adaptive plasticity associated with behavioral training has greatly expanded over the past two decades, but research on maladaptive plasticity occurring with immobilization is scant. A greater understanding of brain structural and functional changes, and the concomitant behavioral effects resulting from limb disuse and unloading has implications for rehabilitation of those immobilized by injury, disease, or even simple inactivity.
Task Progress:	This project was conducted as a part of the “AGBRESA” (Artificial Gravity Bed Rest – European Space Agency) campaign at :envihab in Cologne, Germany. This was a 60 day head down tilt bed rest study with daily centrifugation as a potential countermeasure. One group did not participate in centrifugation, one received 30 minutes per day of continuous centrifugation, and one received 30 minutes per day of intermittent centrifugation in 5 minute sessions. The original aims of the project are: Aim 1) Determine whether 30 minutes per day of supine centrifugation +1Gz at the heart is effective at mitigating declines in brain structure, brain function, sensorimotor and cognitive performance induced by 60 days of 6° head down tilt bed rest in comparison to a control group that is not exposed to AG. ; Aim 2) Determine whether centrifugation is more effective and better tolerated as a countermeasure when it is delivered continuously for 30 minutes or in six, five minute sessions separated by breaks of five minutes each.; Aim 3) Identify predictors of individual differences in bed rest and AG responses. We will examine whether pre bed rest measures of brain structure and brain function are predictive of changes in sensory, motor, and cognitive function occurring in association with bed rest, and whether they also predict AG tolerance. We will combine data collected here with those we acquired for two previously National Space Biomedical Research Institute (NSBRI) funded “predictors” studies (Bloomberg, Mulavara Principal Investigators-PIs) to increase our predictive power. The data were collected at :envihab, with data collection complete in December, 2019. Data transfer from Germany to my laboratory was complete in March, 2020. We had an initial slowing of data analyses due to the COVID pandemic. Therefore, we are still in the midst of manuscript and report preparation. Many of the behavioral measures that we collected show reduced performance from pre to post (or during) bed rest, regardless of the subject group. For example, we found increased time to complete the Functional Mobility Test (a functional obstacle course) and reduced scores on the Sensory Organization Test (SOT) of balance. SOT5 is a condition that taps more into vestibularly mediated control of balance. We also administered some tests while subjects were being spun on the centrifuge or immediately following; performance for these metrics showed group differences. For example, accuracy on the Paced Auditory Serial Addition Test was better for the two artificial gravity groups (who performed the test during centrifugation) than for the control group (who performed the test bedside). The artificial gravity groups also tended to respond faster, supporting that their higher accuracy is not simply a speed-accuracy trade off. It is not clear whether the testing environment might have impacted motivation or attention levels. The continuous artificial gravity group had higher self-reported motion sickness levels immediately following centrifugation (measured on a weekly basis) than the intermittent artificial gravity group. This finding supports a previous report that an intermittent centrifugation dosage is better tolerated by participants than continuous centrifugation. We also recorded the duration of post centrifugation motion illusion, a phenomenon mediated by the semicircular canals in which post-rotatory motion illusion is in the opposite direction to passive rotation. We found that the continuous artificial gravity group reported a longer duration of this motion illusion than the intermittent artificial gravity group. These behavioral group differences for tasks performed during or immediately after centrifugation have potential operational relevance (all behavioral effects described were significant at Benjamini-Hochberg corrected $p < .05$). We collected resting state functional MRI scans to allow us to investigate potential group differences in how brain functional connectivity changes with bed rest. We quantified functional connectivity within multiple brain networks that include the somatosensory, motor, visual, vestibular, and multi-sensory association brain areas. We collapsed across the two artificial gravity groups to compare changes during bed rest with the control group. We found that posterior parietal cortex (a multi-sensory association brain area) showed differential patterns of functional connectivity changes with somatosensory cortex during bed rest for the artificial gravity and control groups (false discovery rate corrected $p < .05$). The control group showed initially increasing connectivity between these regions during bed rest, with the effects returning baseline levels before they exited bed rest. In contrast, the artificial gravity group showed a small but persistent decrease in functional connectivity between these regions during bed rest, which returned to baseline levels by ten days after bed rest. This effect may reflect differential changes in sensory weighting between the groups during bed rest, potentially due to the daily vestibular stimulation as well as somatosensory stimulation (tactile and proprioceptive inputs while the body is loaded) provided by the centrifugation.
Bibliography Type:	Description: (Last Updated: 03/12/2026)

Abstracts for Journals and Proceedings	Tays GD, McGregor HR, Lee JK, Gadd N, Kofman IS, De Dios YE, Wood S, Mulder E, Bloomberg JJ, Mulavara AP, Seidler RD. "Does artificial gravity protect against the effects of a spaceflight analog environment on cognitive and sensorimotor performance?" North Florida Society for Neuroscience, Gainesville, FL (Virtual), February 19, 2021. North Florida Society for Neuroscience meeting, February 2021. Abstracts. , Feb-2021
Abstracts for Journals and Proceedings	Tays GD, McGregor HR, Lee JK, Gadd N, Kofman IS, De Dios YE, Wood S, Mulder E, Bloomberg JJ, Mulavara AP, Seidler RD. "Does artificial gravity protect against the effects of a spaceflight analog environment on cognitive and sensorimotor performance?" 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Abstracts for Journals and Proceedings	McGregor HR, De Dios YE, Gadd NE, Koffman IS, Bloomberg JJ, Wood S, Mulvara AP, Seidler RD. "Effects of Continuous and Intermittent Artificial Gravity of Intracranial Free Water Distribution During Head-down Tilt Bed Rest." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2021
Abstracts for Journals and Proceedings	Tays GD, McGregor HR, Lee JK, Gadd N, Kofman IS, De Dios YE, Wood S, Mulder E, Bloomberg JJ, Mulavara AP, Seidler RD. "Does artificial gravity protect against the effects of a spaceflight analog environment on motor learning?" Abstract was accepted but meeting was cancelled: Society for the Neural Control of Movement Annual Conference, Dubrovnik, Croatia, April 27 – May 1, 2020. Society for the Neural Control of Movement Annual Conference, Dubrovnik, Croatia, April 27 – May 1, 2020. (Meeting cancelled due to Covid.) , Apr-2020
Abstracts for Journals and Proceedings	Tays GD, McGregor HR, Lee JK, Gadd N, Kofman IS, De Dios YE, Wood S, Mulder E, Bloomberg JJ, Mulavara AP, Seidler RD. "Does artificial gravity interact with head-down tilt bedrest to have an effect on motor learning?" University of Florida, North Central Florida Society for Neuroscience, Gainesville, FL, January 31, 2020. Abstracts. University of Florida, North Central Florida Society for Neuroscience, Gainesville, FL, January 31, 2020. , Jan-2020
Articles in Peer-reviewed Journals	Hupfeld KE, McGregor HR, Reuter-Lorenz PA, Seidler RD. "Microgravity effects on the human brain and behavior: Dysfunction and adaptive plasticity." Neurosci Biobehav Rev. 2021 Mar;122:176-89. Review. https://doi.org/10.1016/j.neubiorev.2020.11.017 ; PMID: 33454290 , Mar-2021